

Mathematical Physics I (Fall 2025): Homework #4 Solution

Due Nov. 7, 2025 (Fri, 23:00pm)

[0.5 pt each, total 5 pts]

1. Boas Chapter 7, Problem 5.9

(Note: For Problems 5.9, you are asked to tackle the problem in two ways — first directly find the Fourier coefficients, then verify your result using the answer given in Problem 5.7.)

- From Eqs.(5.6), (5.9)-(5.10) of Boas Chapter 7, you can find

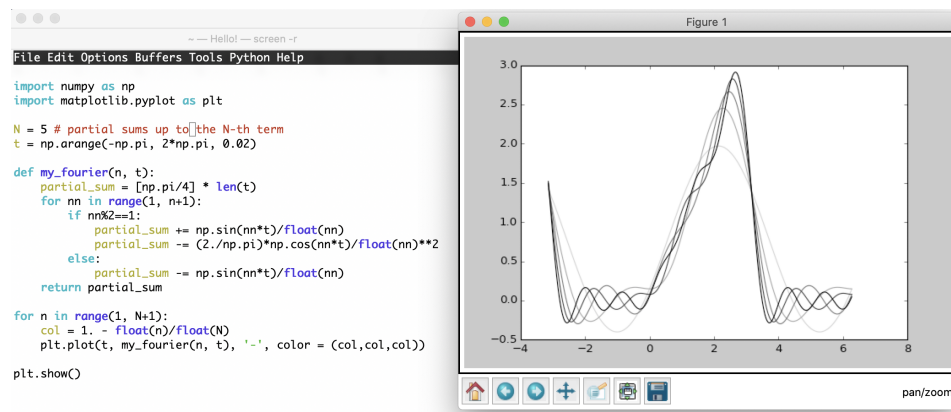
$$\begin{aligned}a_0 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \left\{ - \int_{-\pi}^0 x dx + \int_0^{\pi} x dx \right\} = \pi, \\a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{1}{\pi} \left\{ - \int_{-\pi}^0 x \cos nx dx + \int_0^{\pi} x \cos nx dx \right\} \\&= \frac{1}{\pi} \left\{ - \left[\frac{x \sin nx}{n} \right]_{-\pi}^0 - \left[\frac{\cos nx}{n^2} \right]_{-\pi}^0 + \left[\frac{x \sin nx}{n} \right]_0^{\pi} + \left[\frac{\cos nx}{n^2} \right]_0^{\pi} \right\} \\&= \frac{2}{\pi} \left\{ \frac{(-1)^n - 1}{n^2} \right\} = \begin{cases} 0, & \text{if } n \text{ even,} \\ -\frac{4}{n^2\pi}, & \text{if } n \text{ odd,} \end{cases} \\b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx = \frac{1}{\pi} \left\{ - \int_{-\pi}^0 x \sin nx dx + \int_0^{\pi} x \sin nx dx \right\} \\&= \frac{1}{\pi} \left\{ - \left[-\frac{x \cos nx}{n} \right]_{-\pi}^0 - \left[\frac{\sin nx}{n^2} \right]_{-\pi}^0 + \left[-\frac{x \cos nx}{n} \right]_0^{\pi} + \left[\frac{\sin nx}{n^2} \right]_0^{\pi} \right\} = 0\end{aligned}$$

Since the function $f(x)$ is even, it was expected that all $b_n = 0$. One can quickly verify this Fourier expansion using the answer $g(x)$ to Problem 5.7, as $f(x)$ can be written as $g(x) + g(-x)$.

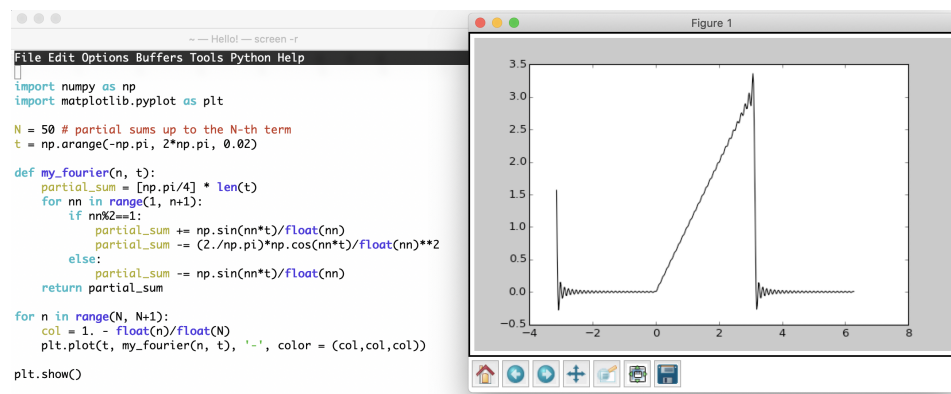
2. Boas Chapter 7, Problem 6.12 (for Problem 5.7)

(Note: For Problems 6.12 and 7.7, you will first have to explicitly work out Problem 5.7. Then for Problem 6.12, you may consider writing a simple script in MATLAB or in python to automatically plot several partial sums for different values of n 's. However, creating such an automated script is not required to receive full credit.)

- An example of an automated plotting script ($n = 1$ to 5) is as follows:



By pushing n to 50, a very good approximation of the given $f(x)$ is acquired:



- You are, however, not required to come up with an automated script like above. You may simply use e.g., WolframAlpha to create multiple plots in a row. For example, the black line ($n = 5$) in the first plot above matches what is seen below:



$\pi/4 - (2/\pi) * [\cos(x) + \cos(3x)/3^2 + \cos(5x)/5^2] + [\sin(x) - \sin(2x)/2 + \sin(3x)/3 - \sin(4x)/4 + \sin(5x)/5]$

NATURAL LANGUAGE

MATH INPUT

EXTENDED KEYBOARD

EXAMPLES

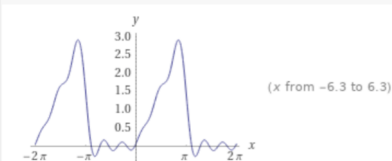
UPLOAD

RANDOM

Input

$$\frac{\pi}{4} - \frac{2}{\pi} \left(\cos(x) + \frac{\cos(3x)}{3^2} + \frac{\cos(5x)}{5^2} \right) + \left(\sin(x) - \frac{1}{2} \sin(2x) + \frac{1}{3} \sin(3x) - \frac{1}{4} \sin(4x) + \frac{1}{5} \sin(5x) \right)$$

Plots



3. Boas Chapter 7, Problem 7.7

- From Eqs.(7.4) and (7.6) of Boas Chapter 7, you can find

$$\begin{aligned} c_0 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{2\pi} \int_0^{\pi} x dx = \frac{\pi}{4}, \\ c_n &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx = \frac{1}{2\pi} \int_0^{\pi} x e^{-inx} dx = \frac{1}{2\pi} \left\{ \left[\frac{x e^{-inx}}{-in} \right]_0^{\pi} + \frac{1}{in} \int_0^{\pi} e^{-inx} dx \right\} \\ &= \frac{1}{2\pi} \left\{ \frac{\pi(-1)^n}{-in} + \frac{(-1)^n - 1}{n^2} \right\} = \begin{cases} \frac{i}{2n}, & \text{if } n \text{ even,} \\ -\frac{i}{2n} - \frac{1}{\pi n^2}, & \text{if } n \text{ odd.} \end{cases} \end{aligned}$$

4. Boas Chapter 7, Problem 8.15(a)

- From Eq.(8.3) of Boas Chapter 7, you can find

$$\begin{aligned} c_0 &= \frac{1}{2} \int_{-1}^1 f(x) dx = \frac{1}{2} \int_{-1}^1 x dx = 0, \\ c_n &= \frac{1}{2} \int_{-1}^1 f(x) e^{-in\pi x} dx = \frac{1}{2} \int_{-1}^1 x e^{-in\pi x} dx = \frac{1}{2} \left\{ \int_0^1 x (e^{-in\pi x} - e^{in\pi x}) dx \right\} \\ &= -i \int_0^1 x \sin n\pi x dx = -i \left\{ - \left[\frac{x \cos n\pi x}{n\pi} \right]_0^1 + \frac{1}{n\pi} \int_0^1 \cos n\pi x dx \right\} = \frac{i \cos n\pi}{n\pi} = \frac{i(-1)^n}{n\pi} \end{aligned}$$

One can then easily show that

$$f(x) = \frac{1}{\pi} \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} \frac{i(-1)^n e^{in\pi x}}{n} = \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{i(-1)^n \{e^{in\pi x} - e^{-in\pi x}\}}{n} = \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n+1} \sin n\pi x}{n}.$$

5. Boas Chapter 7, Problem 11.8

(Note: For Problem 11.8, you are asked to first work out Problem 9.10.)

- From Eqs.(5.6), (5.9)-(5.10) or (9.5) of Boas Chapter 7, you can find $b_n = 0$,

$$a_0 = \frac{2}{\pi} \int_{-\pi/2}^{\pi/2} |x| dx = \frac{4}{\pi} \int_0^{\pi/2} x dx = \frac{\pi}{2},$$

and

$$\begin{aligned} a_n &= \frac{4}{\pi} \int_0^{\pi/2} x \cos 2nx dx = \frac{4}{\pi} \left\{ \left[\frac{x \sin 2nx}{2n} \right]_0^{\pi/2} - \frac{1}{2n} \int_0^{\pi/2} \sin 2nx dx \right\} \\ &= \frac{(-1)^n - 1}{n^2 \pi} = \begin{cases} 0, & \text{if } n \text{ even,} \\ -\frac{2}{n^2 \pi}, & \text{if } n \text{ odd.} \end{cases} \end{aligned}$$

- Then from Parseval's theorem in Eq.(11.4) of Boas Chapter 7, you find

$$\frac{1}{\pi} \int_{-\pi/2}^{\pi/2} |x|^2 dx = \frac{\pi^2}{12} = \left(\frac{1}{2} a_0 \right)^2 + \frac{1}{2} \sum_{n=1}^{\infty} a_n^2 = \frac{\pi^2}{16} + \frac{2}{\pi^2} \sum_{\substack{n=1 \\ n \text{ odd}}}^{\infty} \frac{1}{n^4} \rightarrow \sum_{\substack{n=1 \\ n \text{ odd}}}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{96}.$$

6. Boas Chapter 7, Problem 12.15

(Note: For Problem 12.15, you are asked to first work out Problem 12.9.)

- From Eq.(12.15) of Boas Chapter 7, you can find

$$\begin{aligned}
 g_c(\alpha) &= \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \cos \alpha x \, dx = \sqrt{\frac{2}{\pi}} \int_0^a (-2x + 2a) \cos \alpha x \, dx \\
 &= -\sqrt{\frac{2}{\pi}} \int_0^a 2x \cos \alpha x \, dx + \sqrt{\frac{2}{\pi}} \left[\frac{2a \sin \alpha x}{\alpha} \right]_0^a \\
 &= -\sqrt{\frac{2}{\pi}} \left[\frac{2x \sin \alpha x}{\alpha} \right]_0^a + \sqrt{\frac{2}{\pi}} \int_0^a \frac{2 \sin \alpha x}{\alpha} \, dx + \sqrt{\frac{2}{\pi}} \frac{2a \sin \alpha a}{\alpha} \\
 &= 2\sqrt{\frac{2}{\pi}} \left[-\frac{\cos \alpha x}{\alpha^2} \right]_0^a = 2\sqrt{\frac{2}{\pi}} \cdot \frac{1 - \cos \alpha a}{\alpha^2},
 \end{aligned}$$

which gives the Fourier cosine integral as

$$f_c(x) = \sqrt{\frac{2}{\pi}} \int_0^\infty g_c(\alpha) \cos \alpha x \, d\alpha = \frac{4}{\pi} \int_0^\infty \frac{1 - \cos \alpha a}{\alpha^2} \cos \alpha x \, d\alpha.$$

- Alternatively, from Eq.(12.2) of Boas Chapter 7, you find

$$\begin{aligned}
 g(\alpha) &= \frac{1}{2\pi} \int_{-\infty}^\infty f(x) e^{-i\alpha x} \, dx = \frac{1}{2\pi} \left\{ \int_{-a}^0 (2x + 2a) e^{-i\alpha x} \, dx + \int_0^a (-2x + 2a) e^{-i\alpha x} \, dx \right\} \\
 &= \frac{1}{2\pi} \left\{ \int_0^a (-2x + 2a) e^{i\alpha x} \, dx + \int_0^a (-2x + 2a) e^{-i\alpha x} \, dx \right\} \\
 &= \frac{1}{\pi} \int_0^a (-2x + 2a) \cos \alpha x \, dx
 \end{aligned}$$

which differs from $g_c(\alpha)$ above only by a constant prefactor, thus giving

$$g(\alpha) = \frac{2}{\pi} \cdot \frac{1 - \cos \alpha a}{\alpha^2}.$$

Therefore, the Fourier exponential integral becomes

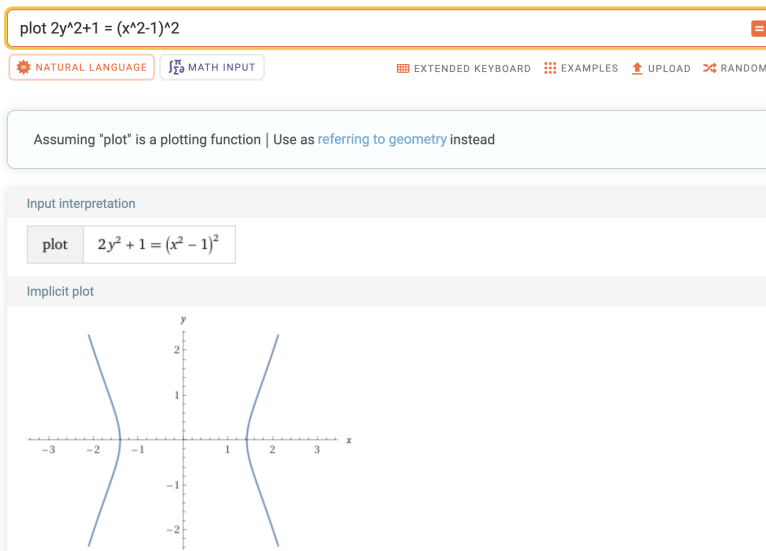
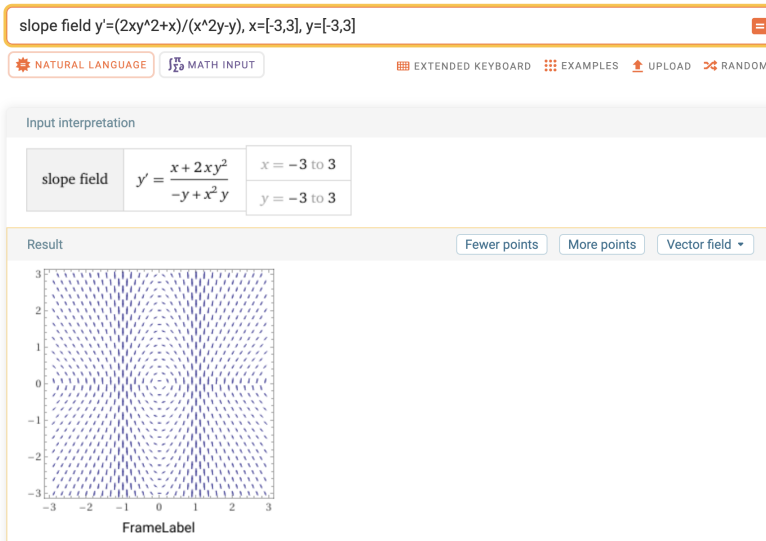
$$f(x) = \int_{-\infty}^\infty g(\alpha) e^{i\alpha x} \, d\alpha = \frac{2}{\pi} \int_{-\infty}^\infty \frac{1 - \cos \alpha a}{\alpha^2} e^{i\alpha x} \, d\alpha = \frac{4}{\pi} \int_0^\infty \frac{1 - \cos \alpha a}{\alpha^2} \cos \alpha x \, d\alpha,$$

which is identical to $f_c(x)$ found above.

7. Boas Chapter 8, Problem 2.6

(Note: For Problem 2.6, read the instruction in the textbook carefully; that is, you have been asked to plot a slope field and the solution curve with a computer. To learn more about the “slope field”, you may review Boas Chapter 8, Section 1.)

- $(x^2 y - y) dy = (2xy^2 + x) dx \rightarrow \frac{y dy}{2y^2 + 1} = \frac{x dx}{x^2 - 1} \rightarrow \frac{1}{4} \ln(2y^2 + 1) = \frac{1}{2} \ln(x^2 - 1) + C \rightarrow 2y^2 + 1 = C' \cdot (x^2 - 1)^2 \rightarrow (x, y) = (\sqrt{2}, 0)$ gives $C' = 1$.

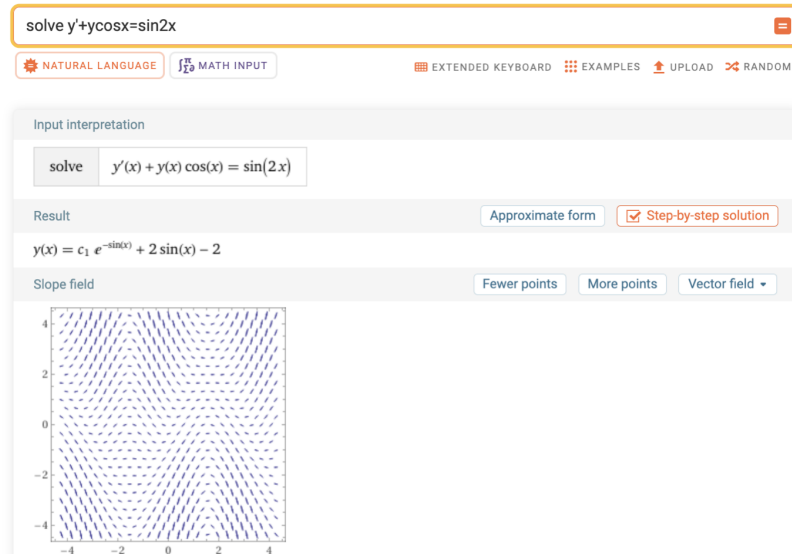


8. Boas Chapter 8, Problem 3.11

(Note: For Problem 3.11, again, read the instruction in the textbook carefully.)

• From Eqs.(3.4) and (3.9) of Boas Chapter 8, $I = \int P dx = \int \cos x dx = \sin x \rightarrow y = e^{-I} \int Q e^I dx + C e^{-I} = e^{-\sin x} \int \sin 2x \cdot e^{\sin x} dx + C e^{-\sin x}$.

• Then, we change the variable to get $\int \sin 2x \cdot e^{\sin x} dx = 2 \int \sin x \cos x \cdot e^{\sin x} dx = 2 \int u e^u du = 2(u e^u - e^u) = 2e^{\sin x}(\sin x - 1)$. Therefore, $y = e^{-\sin x} \int \sin 2x \cdot e^{\sin x} dx + C e^{-\sin x} = 2(\sin x - 1) + C e^{-\sin x}$.



9. Here we study the Fourier series expansion of an example of an impulse-like function, and examine its behavior.

(a) Consider an impulse-like function of width $\frac{1}{N}$, given in the interval $-\pi < x < \pi$:

$$\delta_N(x) = \begin{cases} N, & \text{if } |x| < \frac{1}{2N}, \\ 0, & \text{if } |x| > \frac{1}{2N}. \end{cases}$$

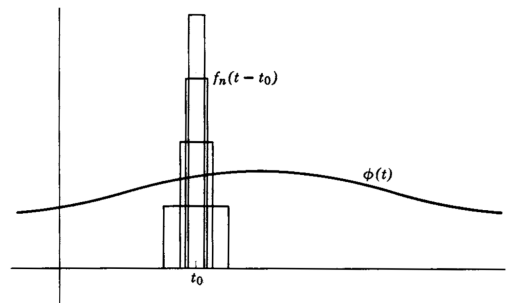
Following the worked Example in Boas Chapter 7, Section 9 (or Problems 9.17-9.22 in Boas Chapter 7), expand $\delta_N(x)$ in a Fourier cosine series.

(b) What does your Fourier series from (a) converge to in the limit as $N \rightarrow \infty$?

(c) Show that for any function $f(x)$ that is finite in the interval $[-\pi, \pi]$ and continuous at $x = 0$,

$$\lim_{N \rightarrow \infty} \int_{-\pi}^{\pi} f(x) [\text{Fourier series expansion of } \delta_N(x)] dx = f(0),$$

which resembles an important property of what we later define as the Dirac delta function, $\delta(x)$.



[Adapted from Boas, Chapter 8.11]

(Note: You may find it useful to preview Eq.(11.6) in Boas Chapter 8, along with the accompanying discussion, which illustrates this property of the Dirac delta function.)

- (a) For even function f_c of period 2π , from Eqs.(5.6), (5.9)-(5.10) or (9.5), you can find $b_n = 0$,

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} \delta_N(x) dx = \frac{2}{\pi} \int_0^{\frac{1}{2N}} N dx = \frac{1}{\pi},$$

and

$$a_n = \frac{2}{\pi} \int_0^{\frac{1}{2N}} \delta_N(x) \cos nx dx = \frac{2}{\pi} \int_0^{\frac{1}{2N}} N \cos nx dx = \frac{2N}{\pi} \left[\frac{\sin nx}{n} \right]_0^{\frac{1}{2N}} = \frac{1}{\pi} \cdot \frac{\sin(n/2N)}{n/2N}.$$

Therefore,

$$\delta_N(x) = \frac{1}{2\pi} + \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{\sin(n/2N)}{n/2N} \cos nx.$$

- (c) Inserting the Fourier expansion of $\delta_N(x)$ from (a),

$$\begin{aligned} \lim_{N \rightarrow \infty} \int_{-\pi}^{\pi} f(x) [\text{Fourier expansion of } \delta_N(x)] dx &= \lim_{N \rightarrow \infty} \int_{-\pi}^{\pi} f(x) \left[\frac{1}{2\pi} + \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{\sin(n/2N)}{n/2N} \cos nx \right] dx \\ &= \int_{-\pi}^{\pi} f(x) \left[\frac{1}{2\pi} + \frac{1}{\pi} \sum_{n=1}^{\infty} \cos nx \right] dx = \frac{1}{2} \cdot \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx + \sum_{n=1}^{\infty} \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx \\ &= \frac{a'_0}{2} + \sum_{n=1}^{\infty} a'_n, \end{aligned}$$

where a'_n is the Fourier coefficients of $f(x)$. This simply is $\left[\frac{a'_0}{2} + \sum_{n=1}^{\infty} a'_n \cos nx \right]_{x=0} = f(0)$.

10. Let us use the Fourier transform to appreciate the meaning of the Heisenberg uncertainty principle in quantum mechanics.

(a) Imagine an infinite wave train $\sin \omega_0 t$ clipped by shutters to maintain only N cycles of the original waveform:

$$f(t) = \begin{cases} \sin \omega_0 t, & \text{if } |\omega_0 t| < N\pi, \\ 0, & \text{if } |\omega_0 t| > N\pi. \end{cases}$$

Find the amplitude function of the Fourier (exponential) transform, $g(\omega)$. Since the prefactor may depend on the exact definition of the transform, do not worry too much about it.

(b) Find the amplitude function of the Fourier sine transform, $g_s(\omega)$. Again, since the prefactor may depend on the exact definition of the transform, do not worry too much about it.

(c) Show that, in the special case of $N = \frac{1}{2}$ and $\omega_0 = 1$, $g_s(\omega)$ becomes

$$g_{s,1}(\omega) = \sqrt{\frac{2}{\pi}} \frac{\omega \cos\left(\frac{\omega\pi}{2}\right)}{1 - \omega^2}.$$

(d) Now consider the limit of $\omega_0 \gg 1$ and $\omega \approx \omega_0$. Show that $g_s(\omega)$ is approximately equal to

$$g_{s,2}(\omega) = \frac{1}{\sqrt{2\pi}} \frac{\sin\left[(\omega_0 - \omega) \frac{N\pi}{\omega_0}\right]}{\omega_0 - \omega}.$$

Sketch or computer plot this function for $N = 1, 3, 5, 10$ and an arbitrary ω_0 . Notice that for $N \gg 1$, $g_{s,2}(\omega)$ may be interpreted as proportional to what we later define as the Dirac delta function, $\delta(\omega - \omega_0)$, in Boas Chapter 8, Section 11.

(Note: You do not need to provide a mathematically rigorous proof that $g_{s,2}(\omega)$ indeed becomes proportional to $\delta(\omega - \omega_0)$ as $N \rightarrow \infty$. For now, observe the shape of $g_{s,2}(\omega)$ as you vary N , and based on your observation simply argue that $\lim_{\omega \rightarrow \omega_0} g_{s,2}(\omega) \rightarrow \infty$ as $N \rightarrow \infty$.)

(e) Show that the first zeros of $g_{s,2}(\omega)$ from $\omega = \omega_0$ are at $\omega = \omega_0 \pm \Delta\omega = \omega_0 \pm \frac{\omega_0}{N}$. Justify that $\Delta\omega = \frac{\omega_0}{N}$ could be a good measure of the spread (or uncertainty) in frequency of our clipped wave train. Then, establish the inverse relationship between the wave train's pulse length ($N\pi$) and the frequency spread ($\Delta\omega$). Finally, using the relationship along with the assumed *wave nature of matter*, explain the uncertainty principle of quantum mechanics.

(Note: You may want to briefly review the textbooks in quantum mechanics such as Griffiths & Schroeter. Note that the inverse relationship found here is a fundamental property of the finite wave train, and has little to do with any additional *ad hoc* postulates in quantum mechanics. If needed, you must reference your sources appropriately with a proper citation convention, but your answer must still be your own work in your own words. To access the electronic resources — e.g., academic journals — off-campus via SNU library's proxy service, see <http://library.snu.ac.kr/using/proxy>.)

- (a) From Eq.(12.2) of Boas Chapter 7, you find

$$\begin{aligned} g(\omega) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt = \frac{1}{2\pi} \int_{-\frac{N\pi}{\omega_0}}^{\frac{N\pi}{\omega_0}} \sin \omega_0 t e^{-i\omega t} dt = \frac{1}{2\pi} \int_{-\frac{N\pi}{\omega_0}}^{\frac{N\pi}{\omega_0}} \frac{e^{i\omega_0 t} - e^{-i\omega_0 t}}{2i} \cdot e^{-i\omega t} dt \\ &= \frac{1}{2\pi i} \left[\frac{e^{i(\omega_0 - \omega)\frac{N\pi}{\omega_0}} - e^{-i(\omega_0 - \omega)\frac{N\pi}{\omega_0}}}{2i(\omega_0 - \omega)} - \frac{e^{i(\omega_0 + \omega)\frac{N\pi}{\omega_0}} - e^{-i(\omega_0 + \omega)\frac{N\pi}{\omega_0}}}{2i(\omega_0 + \omega)} \right] \\ &= \frac{1}{2\pi i} \left[\frac{\sin[(\omega_0 - \omega)\frac{N\pi}{\omega_0}]}{\omega_0 - \omega} - \frac{\sin[(\omega_0 + \omega)\frac{N\pi}{\omega_0}]}{\omega_0 + \omega} \right]. \end{aligned}$$

- (b) From Eq.(12.14) of Boas Chapter 7, you can find

$$g_s(\omega) = \sqrt{\frac{2}{\pi}} \int_0^{\frac{N\pi}{\omega_0}} \sin \omega_0 t \sin \omega t dt = \frac{1}{\sqrt{2\pi}} \left[\frac{\sin[(\omega_0 - \omega)\frac{N\pi}{\omega_0}]}{\omega_0 - \omega} - \frac{\sin[(\omega_0 + \omega)\frac{N\pi}{\omega_0}]}{\omega_0 + \omega} \right].$$

- (d) $\lim_{\omega \rightarrow \omega_0} g_{s,2}(\omega) = \frac{1}{\sqrt{2\pi}} \frac{N\pi}{\omega_0} \rightarrow \infty$ as $N \rightarrow \infty$. Computer plots of $g_{s,2}(\omega)$ for $N = 1$ and 5 can be found below.

$y = \sin((1-w) \cdot \pi / 1) / (1-w)$, w from -9 to 11, y from -1 to 4

 NATURAL LANGUAGE

 MATH INPUT

 EXTENDED KEYBOARD

 EXAMPLES

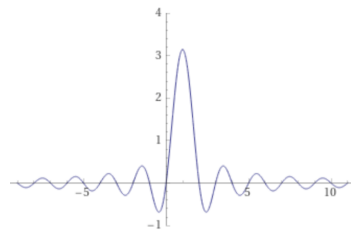
 UPLOAD

 RANDOM

Input interpretation

plot	$y = \frac{\sin((1-w) \times 1 \times \frac{\pi}{1})}{1-w}$	$w = -9$ to 11
		$y = -1$ to 4

Plot



$y = \sin((1-w) \cdot 5 \cdot \pi / 1) / (1-w)$, w from -9 to 11, y from -4 to 16

 NATURAL LANGUAGE

 MATH INPUT

 EXTENDED KEYBOARD

 EXAMPLES

 UPLOAD

 RANDOM

Input interpretation

plot	$y = \frac{\sin((1-w) \times 5 \times \frac{\pi}{1})}{1-w}$	$w = -9$ to 11
		$y = -4$ to 16

Plot

